AAYUSH BHATTARAI

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PROFILE

Final-year Information Technology Engineering student at South-Eastern Finland University of Applied Sciences (Xamk) with a 3.97 weighted average and a completed thesis on the design, implementation, and security of a hierarchical Cisco enterprise network. Hands-on experience with data center environments, routing and switching, Layer 2 security, and Linux server administration. Comfortable with physical hardware work, shift-based operations, and cross-team collaboration. Seeking to contribute to Nebius' Mäntsälä data center as an Early Talent IT Datacenter Technician.

EDUCATION

Bachelor of Engineering, Information Technology | *South-Eastern Finland University of Applied Sciences (Xamk), Mikkeli*
Aug 2021 – 2026

- Weighted average: 3.97 / 5 • 255 / 240 credits completed
- Thesis: "Design, Implementation, and Security of a Hierarchical Enterprise Network Using Cisco and Open-Source Technologies" — Grade 4
- Relevant coursework: Data center environments, Routing and switching essentials, Scaling networks, Advanced networking, Information and network security, Operating systems and server environments, IoT technologies, Python programming and data processing

TECHNICAL SKILLS

- **Data Center & Infrastructure:** Server hardware, rack-mounted equipment, fiber optic and copper cabling, patch panels, structured cabling, hardware diagnostics and replacement
- **Networking:** Cisco routing & switching (CCNAv7), VLANs, inter-VLAN routing, EtherChannel, DHCP/DNS, 802.1X, port security, DHCP snooping, TCP/IP, subnetting
- **Systems & OS:** Linux (Ubuntu, Debian), Windows Server, PowerShell, Bash, virtualization, Raspberry Pi, Pi-hole
- **Cloud & DevOps:** Microsoft Azure, AWS (Academy Cloud Foundations graduate), Git/GitHub, GitLab, Jenkins, CI/CD basics
- **Programming & Automation:** Python (RESTCONF/NETCONF network automation), C++, JavaScript/TypeScript, Robot Framework, Selenium
- **Languages:** English (fluent), Finnish (intermediate — completed Finnish 1, 2, and Finnish for Foreigners 3), Nepali (native), Hindi (fluent)

PROJECTS

Hierarchical Enterprise Network — Bachelor's Thesis (2025 – 2026)

- Designed and implemented a three-layer (core / distribution / access) Cisco-based enterprise network, including VLAN segmentation, inter-VLAN routing, and EtherChannel link aggregation for redundancy.
- Deployed DHCP and DNS services using Pi-hole on a Linux host; integrated 802.1X authentication and IPS for endpoint and perimeter security.
- Hardened Layer 2 with DHCP snooping, port security, and dynamic ARP inspection; documented the full configuration, topology, and security posture. Final grade: 4/5.

IoT-Based Smart Home Network

- Built a segmented home-lab network with Raspberry Pi acting as gateway, DNS sinkhole, and monitoring node; configured secure communication between IoT devices and applied network-level access controls.

Network Automation with RESTCONF / NETCONF

- Automated configuration of Cisco devices using Python with RESTCONF and NETCONF APIs, reducing manual CLI work and producing repeatable, version-controlled network configs.

Enterprise Network Security Assessment

- Performed a vulnerability analysis on a simulated enterprise network, identified weaknesses in access control and segmentation, and implemented mitigations validated through testing.

EXPERIENCE

Full-Stack Web Developer Intern | *Metatavu Oy* Feb 2025 – Jul 2025

- Built and maintained production web applications using React, TypeScript, and Node.js as part of an Agile / Scrum team.
- Integrated REST APIs and databases; collaborated with designers and backend engineers; worked with Git-based workflows and code review.
- Gained experience with containerized development environments, CI/CD pipelines, and incident triage on shared infrastructure.

Shift Manager | *Hesburger Joutsensilta* Jul 2025 – Present

- Lead shifts in a fast-paced 24/7-style environment: supervise staff, manage operations, and respond to issues in real time — directly relevant to shift-based data center operations.

Assistant Restaurant Manager | *Hesburger Mikkeli Toripaviljonki* Oct 2022 – Jun 2025

- Acted as substitute restaurant manager; handled staff scheduling, training, inventory, and procedural compliance across rotating shifts.
- Developed strong ownership, safety-awareness, and cross-team communication — transferable to data center floor operations.

CERTIFICATIONS & COURSES

- Cisco Networking Academy: Networking Essentials; CCNAv7 Switching, Routing & Wireless Essentials; CCNAv7 Enterprise Networking, Security & Automation
- Cisco Emerging Technologies Workshop: Model Driven Programmability
- AWS Academy Cloud Foundations — Graduate
- The Complete 2024 Web Development Bootcamp — Certificate of Completion
- Advanced PowerShell (cross-institutional study, Metropolia UAS, 2025)

ADDITIONAL INFORMATION

- Work authorization: Resident in Finland with right to work; no sponsorship required for the Mäntsälä, Finland role.
- Availability: Full-time, 8+ weeks, starting June 2026. Comfortable with shift-based schedules.